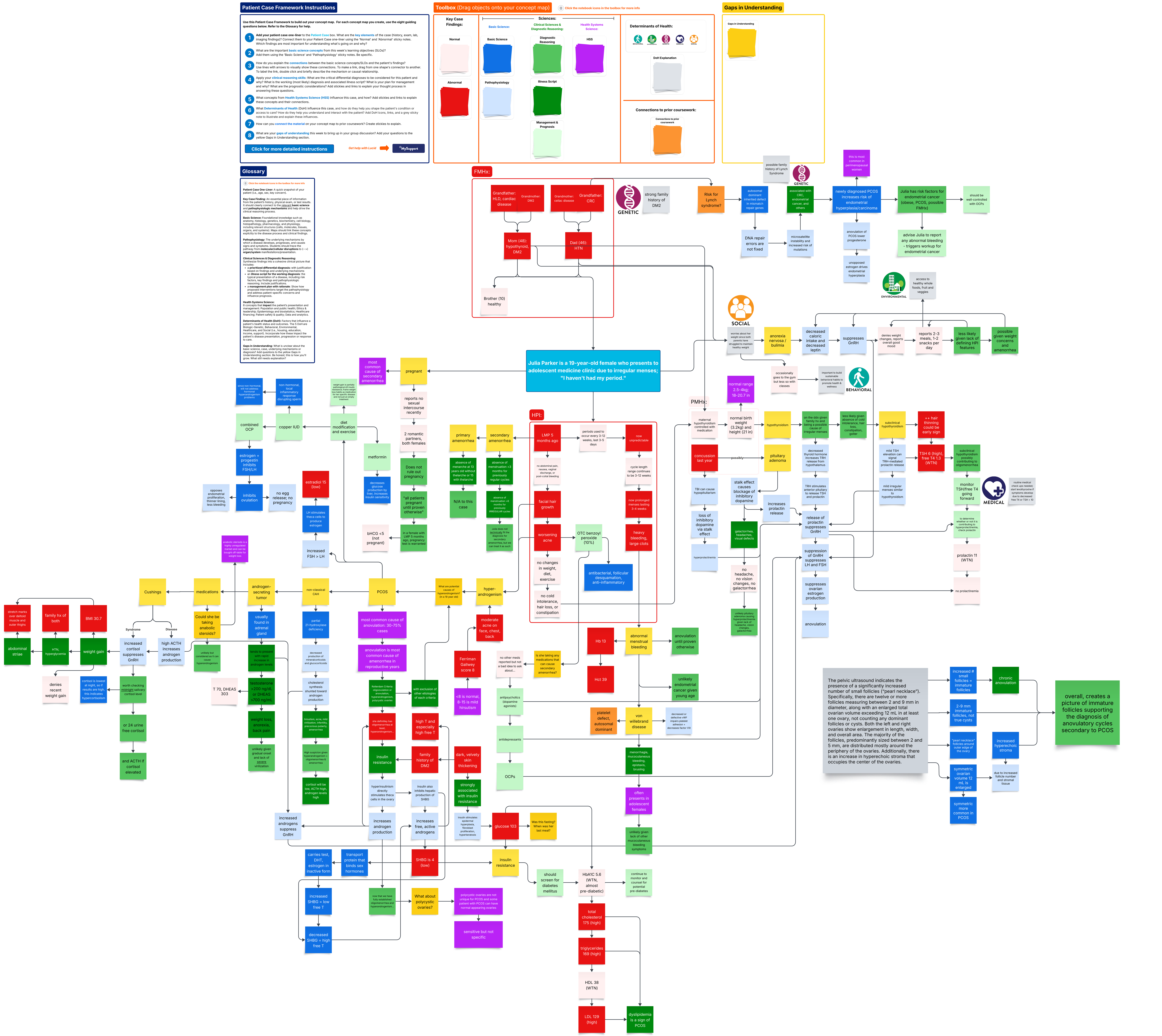

Patient Case Framework Instructions
Toolbox (Drag objects onto your concept map)
Gaps in Understanding
Key Case Findings:
Normal
Abnormal
Sciences:
Basic Science:
Clinical Sciences & Diagnostic Reasoning:
Health Systems Science:
Basic Science
Pathophysiology
Diagnostic Reasoning
Illness Script
Management & Prognosis
HSS
Determinants of Health:
DoH Explanation
Connections to prior coursework:
Click for more detailed instructions
Glossary
FMHx:
Grandfather: HLD, cardiac disease
Grandmother: DM2
Grandmother: celiac disease
Grandfather: CRC
Mom (48) hypothyroid, DM2
Dad (46): HTN
Brother (10) healthy
GENETIC
strong family history of DM2
Risk for Lynch syndrome?
newly diagnosed PCOS increases risk of endometrial hyperplasia/carcinoma
Julia has risk factors for endometrial cancer (obese, PCOS, possible FMHx)
advise Julia to report any abnormal bleeding - triggers workup for endometrial cancer
DNA repair errors are not fixed
SOCIAL
anorexia nervosa / bulimia
decreased caloric intake and decreased leptin
suppresses GnRH
reports 2-3 meals, 1-2 snacks per day
less likely given lack of defining HPI features
possible given weight concerns and amenorrhea
ENVIRONMENTAL
BEHAVIORAL
Julia Parker is a 19-year-old female who presents to adolescent medicine clinic due to irregular menses; "I haven't had my period."
pregnant
most common cause of secondary amenorrhea
reports no sexual intercourse recently
2 romantic partners, both females
Does not rule out pregnancy
"all patients pregnant until proven otherwise"
bHCG <5 (not pregnant)
primary amenorrhea
secondary amenorrhea
N/A to this case
HPI:
LMP 5 months ago
now unpredictable
facial hair growth
worsening acne
no changes in weight, diet, exercise
no cold intolerance, hair loss, or constipation
heavy bleeding, large clots
OTC benzoyl peroxide (10%)
antibacterial, follicular desquamation, anti-inflammatory
PMHx:
normal range 2.5-4kg; 18-20.7 in
concussion last year
pituitary adenoma
hypothyroidism
subclinical hypothyroidism
++ hair thinning could be early sign
TSH 6 (high), free T4 1.3 (WTN)
monitor TSH/free T4 going forward
MEDICAL
prolactin 11 (WTN)
no prolactinemia
release of prolactin suppresses GnRH
suppression of GnRH suppresses LH and FSH
suppresses ovarian estrogen production
anovulation
increases prolactin release
combined OCP
copper IUD
diet modification and exercise
metformin
estrogen + progestin inhibits FSH/LH
inhibits ovulation
no egg release; no pregnancy
estradiol 15 (low)
increased FSH > LH
PCOS
hyper-androgenism
androgen-secreting tumor
non-classical CAH
medications
Cushings
Could she be taking anabolic steroids?
stretch marks
family hx of both
BMI 30.7
abdominal striae
weight gain
increased cortisol suppresses GnRH
high ACTH increases androgen production
denies recent weight gain
or 24 urine free cortisol
and ACTH if cortisol elevated
T 70, DHEAS 303
usually found in adrenal gland
most common cause of anovulation: 30-75% cases
anovulation is most common cause of amenorrhea in reproductive years
moderate acne on face, chest, back
Ferriman Gallwey score 8
<8 is normal, 8-15 is mild hirsutism
insulin resistance
family history of DM2
dark, velvety skin thickening
strongly associated with insulin resistance
high T and especially high free T
increases androgen production
increases free, active androgens
increased androgens suppress GnRH
glucose 103
Was this fasting? When was her last meal?
SHBG is 4 (low)
insulin resistance
should screen for diabetes mellitus
HbA1c 5.6 (WTN, almost pre-diabetic)
total cholesterol 175 (high)
triglycerides 169 (high)
HDL 38 (WTN)
LDL 129 (high)
dyslipidemia is a sign of PCOS
carries test, DHT, estrogen in inactive form
transport protein that binds sex hormones
increased SHBG = low free T
decreased SHBG = high free T
What about polycystic ovaries?
sensitive but not specific
Hb 13
Hct 39
abnormal menstrual bleeding
anovulation until proven otherwise
unlikely endometrial cancer given young age
platelet defect, autosomal dominant
von willebrand disease
often presents in adolescent females
antipsychotics (dopamine agonists)
antidepressants
OCPs
The pelvic ultrasound indicates the presence of a significantly increased number of small follicles ("pearl necklace"). Specifically, there are twelve or more follicles measuring between 2 and 9 mm in diameter, along with an enlarged total ovarian volume exceeding 12 mL in at least one ovary, not counting any dominant follicles or cysts. Both the left and right ovaries show enlargement in length, width, and overall area. The majority of the follicles, predominantly sized between 2 and 5 mm, are distributed mostly around the periphery of the ovaries. Additionally, there is an increase in hyperechoic stroma that occupies the center of the ovaries.
increased # small follicles = immature follicles
2-9 mm immature follicles, not true cysts
"pearl necklace" follicles around outer edge of the ovary
symmetric ovarian volume 12 mL is enlarged
symmetric more common in PCOS
chronic anovulation
increased hyperechoic stroma
overall, creates a picture of immature follicles supporting the diagnosis of anovulatory cycles secondary to PCOS